

# Project Proposal – Internship

Version: 7  
Period: Spring 2023



## Goal and use of this document

The proposal is used to describe a practical research internship which is carried out in semester 5 of the study program. Through this form the student requests approval for the assignment from the internship coordinator. This document is also used to receive feedback from client and coordinator and should lead to all three parties having one single view of the assignment. The student is responsible for writing the content, based on input of the organization and feedback of the coordinator. This document may be written in Dutch.

## Student details

Student number : 4853121  
First name + Family name : Marian-Metin Constantin  
Location : Eindhoven  
Profile Semester 3 : GenAi  
Specialisation Semester 4 : MachineLearning  
Internship choice : GenAi/MachineLearning  
Dutch-speaking? : No

## Internship period

Start date : 31/08/2026 (official start date: Monday FHICT-week 1)  
End date : 22/01/2027 (official end date: Friday FHICT-week 18)

## Organisation details

Name : Nahayat  
Visiting address : Laarakkerweg 6a  
Zipcode + City + Country : 5061JR, Oisterwijk  
Phone : 0855808721  
Website : <https://www.nahayat.io/>  
Own Company ("Eigen bedrijf")? : No

## Company mentor

(The person who guides the student on a regular basis)

First name + Family name : Luc Tortike  
Department :  
Position : AI/Cloud Engineer | Senior Consultant  
Background (highest education) : MSc Data Science & AI  
Background URL (e.g. LinkedIn) : <https://www.linkedin.com/in/luctortike/>  
Phone : 0639881390  
Email : [lmortike@nahayat.io](mailto:lmortike@nahayat.io)  
Assignment in ASAM? : No If yes, link to ASAM: <copy ASAM URL>

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## 1. Context & Problem/Opportunity

*Below the organization and its activities and structure are described. It must become clear in which part of the organization and activities the problem or opportunity lies and what the reason is for this assignment.*

### 1.1 Context & Background

*The organization is introduced by describing its main activities and processes. The relevant (recent) developments within the company and or related field of interest (e.g. IT) are described also.*

Nahayat is an AI consultancy in Oisterwijk, founded by Robin Damen, focused on practical AI for SMBs. Besides giving advise, we build and run the agents ourselves, using open-source components so clients keep control over cost and stack, with no vendor lock-in. We host everything through our sister company Virtual Computing's Dutch datacenters which matters especially with regards to the AVG/GDPR and EU AI act.

Our main project is Alex, an AI agent handling Virtual Computing's IT helpdesk end-to-end. However, there are multiple agentic projects running.

Thus far, it has been difficult to self host open-source LLMs, as they often require heavy computing power. However, recently there has been an increase in promising open source LLMs, also built for agentic tasks. While we have done some experimentation, we would like to get a better view of the capabilities, possibilities, and limitations of these open source LLMs. As we would love to be able to host LLMs in house, wherever possible.

### 1.2 Current situation & Stakeholders

*The current situation regarding the activities, problems (or opportunity) is described. The relevant stakeholders within the organization are provide with an explanation why they (possibly) are stakeholders. The scope of the project must become clear at this point.*

Our orchestration layer (agents, MCP servers, workflows) is self-built and self-hosted. The models behind it are almost always still cloud APIs (Azure OpenAI, AWS Bedrock/Anthropic). While we have regulations with them with regards to data retention, so we can still handle this in a safe way, it would be even better if we could have some self hosted LLM inference and we would like to get a better view of what we can use this for (internal agentic chats, external agents, smaller tasks, chatting, automations, etc.). Note our hardware availability is limited, so we are mainly interested in what we can achieve and when it would be worth upgrading hardware.

#### Stakeholders:

- Nahayat – the company and its employees
- Luc Tortike – employee of Nahayat, and supervisor for this project
- (indirectly) SMB clients of Nahayat

Scope: internal feasibility research and prototyping

## 1.3 Problem/opportunity description

*The problems/opportunities, possible causes and impact or effects of these problems/opportunities are described. Concrete examples are given and the depth and scope of the situation is made clear. In this section it should become clear what the added value of the internship will be.*

We don't know, in any structured way, which of our real use cases, chat, RAG, agentic tool-use, classification, automation, a local model can already handle, what's realistic on hardware without a strong GPU (or when it is worth investing in a proper GPU, as this is a realistic possibility).

The added value will mainly lie in this analysis and some prototypes to showcase results (that might be used in production if they add value). This helps us understand what tasks we can divert to local models (and which models are most promising for which tasks), when it is worth investing in better hardware / GPUs, and what the current possibilities are. Ideally everything is set up in a way that allows swapping LLM models easily).

## 2. Assignment

### 2.1 Desired project result

*Below it should be clear what the desired result of the internship is. What situation should be reached at the end of the internship?*

#### **Project goal**

*A clear and as concrete as possible description of the intended results and what value they should have for the organization. A (concept) project goal can be formulated and products to be delivered are indicated.*

By the end of the internship, we have a clear comparative view of which open-source models are viable at which hardware tier for our use cases, backed by working prototypes and possibly a clear report plus a concrete answer on if/when hardware investment would be worth it.

Deliverables could be (we can still further discuss this):

- A report or a clear representation of findings

#### **Project IT-deliverable**

*Specifically indicate the IT-products (proof-of-concept) that will be developed. Describe the end product as well as any intermediate products or other deliverables that are already known. For each IT-deliverable, specify the technology(s) to be used.*

Working prototypes (e.g. local agentic system, chat/RAG setup, or other possibilities, it is probably best to discuss this further). Also possible is some form of a determined test suite that can be used to evaluate models on certain tasks, so it is possible to get a clear evaluation of the performance of different models.

Possibilities tech wise:

- Ollama, llama.coo/GGUF for possibly quantized CPU inference, Pythn, Langchain, Supabase, Qdrant, other open source frameworks, etc.

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## 2.2 Research aspects

### Research questions

*As it is important that all results are based upon (also practical) research: Which research questions are going to be relevant in order to achieve the desired project result? Provide (sub)-research questions and possibly a main research question.*

Possible RQ:

“To what extent can locally-hosted open-source LLMs, on limited hardware, be used for chat, retrieval, classification, and agentic tool-use, and when would hardware investment be justified?”

1. Which models, at which hardware levels, run acceptably?
2. How does output quality compare to current commercial SOTA models, for different use cases?
3. What are limitations? Tool calling reliability, context length, speed of token output?
4. Which use cases are viable locally, to what extent?

Etc.

### Research approach

*Regarding the DOT Framework: give a first indication of which research strategies and methods will be used and/or will be the most important ones. Think of the possibilities your internship environment specifically has or lacks and how you can in any case ensure you have enough methods and strategies to use.*

### Research strategies and methods – DOT Framework

For this project, the most important research strategy will be **research through experimentation and prototyping**. Since the main research question focuses on the extent to which locally hosted open-source LLMs can perform different tasks on limited hardware, practical testing will be necessary to determine what is actually feasible. Different models, hardware configurations, and AI use cases will be implemented and tested in the internship environment.

**Literature research** will be used to establish the theoretical and technical foundation of the project. This will include research into locally hosted LLMs, model quantization, inference, retrieval-augmented generation (RAG), AI agents, tool calling, and local AI infrastructure. Existing benchmarks and research can also be used to identify suitable models and evaluation methods before conducting the practical experiments.

The main practical method will be **controlled experimentation and benchmarking**. Different open-source models will be deployed on the available hardware and tested across the intended use cases: chat, retrieval, classification, and agentic tool use. Measurements such as response quality, inference speed, time to first token, tokens per second, memory/VRAM usage, context length, and tool-calling reliability can be collected. This will make it possible to compare models and hardware configurations using measurable criteria rather than subjective impressions.

**Comparative research** will be used to compare the local models with suitable commercial state-of-the-art models. The comparison will focus on the requirements of each use case rather than simply determining which model produces the best general answers. For example, a smaller local model may be sufficient for classification or a specific RAG application even if it performs worse than a commercial model in general conversation.

## 3. Guidance & Expertise

### Coach/Mentor

*What background does your company mentor have? Think of education (Bachelor/Master/Phd) as well as experience in (internship) coaching or mentoring. How many hours of guidance will there be per week and will you be coached?*

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Luc works as an AI/Cloud Engineer & Senior Consultant and studies MSc Data Science & AI at TU/e. He is responsible for building and maintaining most of Nahayat's AI agent infrastructure and related development work, responsible for most of the agentic deployments for the sister company Virtual Computing.

## Expertise / Guidance

*What expertise is available in the company and who has that expertise? What kind of support does the company provide on IT skills related to your domain? Describe the people and/or teams or departments where you are able to go to and who will help and guide you with (IT)-domain related questions, advice and review.*

Lot of IT experiment, and frontier when it comes to agentic AI implementations. As we are based in Virtual Computing's office, it is also possible to go to Virtual Computing's support engineers for assistance, or guidance when it comes to IT related questions.

Guidance and supervision:

Company supervisor/couch is available 3 (three) days per week to provide direct guidance and support through the project. In addition, I can approach senior team members for technical assistance, feedback, question when needed.

## 4. Personal & Professional goals

Which aspects on personal or professional development will you further develop? Focus on soft-skills. Select a realistic three to five aspects to work on. Use previous feedback received from others (e.g. semester coaches and teachers as input).

| Personal goal | Describe what exactly you want to achieve and how to work on this.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Communication | I want to improve how I communicate technical information to different audiences. Because this project involves complex topics such as AI models, hardware, and infrastructure, it is important to be able to explain technical findings clearly and concisely, both to technical colleagues and to people with less technical knowledge. I will work on this by regularly presenting my progress, discussing results with colleagues, and practicing explaining technical concepts in a structured way. |
| Planning      | I want to become more consistent in planning my work and managing my time during a longer independent project. The project contains many possible directions to investigate, so it will be important to prioritize what is relevant to the research question and avoid spending too much time on interesting but less relevant experiments. I will use planning, milestones, and regular progress reviews to keep the project focused.                                                                   |
| Feedback      | I want to become more proactive in asking for feedback instead of waiting until a task or project phase is finished. Feedback from supervisors and colleagues can help me identify problems earlier and improve the quality of my work. I will therefore regularly discuss my approach and results with others and document important feedback so that I can actively apply it in the next iteration.                                                                                                    |
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